

LIAM MCCAUSLAND

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SUMMARY

Computer Science student at the University of Manitoba with experience developing Java applications using object-oriented design, data structures, algorithms, and software engineering principles. Strong problem-solving, communication, and teamwork skills developed through academic projects and professional experience. Seeking opportunities to apply technical skills and gain hands-on software development experience.

EXPERIENCE

Elite Professional – Elite Sports Injury

Aug 2024 – Present

- Managed the schedule of physiotherapist clients, including booking, cancelling, and contacting clients.
- Regularly cleaned and maintained exercise areas, including disinfecting equipment and organizing weights.
- Applied and removed heat packs and TENS machines for clients.
- Processed client payments using TD payment terminals.
- Guided clients through rehabilitation programs designed by physiotherapists.
- Managed patient insurance and coverage programs (Military, WCB, MPI) and provided invoices.
- Closed the business day by charting payments and verifying payment totals.

Zoo Camp Supervisor Volunteer - Assiniboine Park Zoo

Aug 2022

- Collaborated closely with fellow supervisors and instructors to provide attentive care and supervision for children engaged in zoo camp activities.
- Utilized strong social and leadership abilities to make prompt decisions and effectively collaborate with other instructors.
- Applied effective communication techniques to engage with the group of children, ensuring that they had the most enriching and enjoyable experience possible

Stadium Operations General Labourer – Investors Group Field

Jun 2021 – Aug 2022

- Completed event setup and teardown for major events and rentals, including tables, chairs, pipe and drape, stages, and podiums according to floor plans.
- Assisted with stadium maintenance, including groundskeeping, pressure washing, and seasonal startup activities.
- Assisted with field painting, including football and soccer lines and field logos.
- Completed field setup and teardown for football and soccer game days, including nets, goal posts, benches, advertising boards, seats, and shelters.
- Assisted with tailgate area setup, including tents, tables, chairs, and bike racks.

EDUCATION

Collège Sturgeon Heights Collegiate

High School Diploma, International Baccalaureate *Graduated 2023*

University of Manitoba

Faculty of Science – Computer Science *Present*

Relevant Course Work

- COMP 2080 - Algorithms: Design and Implementation

- COMP 2140 - Data Structures: Analysis and Implementation
- COMP 2280 - Introduction to Computer Systems
- COMP 2400 - Programming Paradigms
- COMP 2450 - Software Development I
- COMP 2452 - Software Development II
- COMP 3030 - Automata Theory and Formal Languages
- MATH 1240 - Elementary Discrete Mathematics
- STAT 1150 - Introduction to Statistics and Computing

PROJECTS

Battleship++ (Java)

University of Manitoba – COMP 2450

- Developed a turn-based Battleship variant in Java using object-oriented design principles.
- Implemented ship movement, special abilities, game effects, and custom game logic.
- Designed a modular architecture with separation of concerns between game logic, persistence, and user interface components.
- Built JSON-based save/load functionality for game persistence.
- Created unit tests to verify game behavior and improve reliability.
- Produced UML diagrams, flowcharts, and technical documentation throughout development.

Personal Website (HTML, CSS, JS, React, Chakra UI)

liammccausland.com

- Designed and developed a personal portfolio website showcasing projects, skills, and experience.
- Maintained project documentation and professional online presence.
- Integrated GitHub repositories and project information.
- Features more personal side projects

Roblox Game (Lua)

Technologies: Lua, Roblox Studio

- Recreated Valve's Ricochet as a multiplayer Roblox game titled Rebound.
- Implemented core gameplay mechanics including projectile bouncing physics, player movement interactions, and round-based game flow.
- Developed game systems using Lua scripting, including collision handling, scoring logic, and match state management.
- Iterated on gameplay balance and responsiveness through playtesting and refinement.
- Developed custom disc physics script that acts as the anchor of the game.

Automata Visualizer (JavaScript, JointJS)

Technologies: JavaScript, JointJS, HTML/CSS

- Built an interactive finite automata editor and simulator using JointJS and vanilla JavaScript.
- Implemented dynamic state creation, transition linking, and self-loop rendering with automatic graph layout adjustments.
- Developed real-time graph interaction features including drag-and-drop state editing, edge labeling, and transition editing via keyboard input.
- Designed support for accepting states, start state visualization, and labeled transition systems.
- Implemented custom formatting utilities for mathematical symbols (e.g., Greek letters and subscripts) to support formal automata notation.
- Added full graph editing capabilities including deletion, selection states, and interactive transition creation via context menus.

TECHNICAL SKILLS

Languages

- JavaScript
- HTML/CSS
- C, C++, C#

Software Development

- Object-Oriented Programming
- Unit Testing
- JSON Data Persistence
- UML Design
- Debugging & Refactoring
- SOLID Principles

Tools

- Git
- GitHub
- Microsoft Office 365
- IntelliJ IDEA / VS Code
- Vim

PROFESSIONAL SKILLS

- Communication
- Teamwork
- Problem Solving
- Time Management
- Adaptability